

PROBLEM

1.EMERGENCY MEDICINE DELIVERY USING AI DRONES

Problem Statement

Develop an AI-based drone navigation system that delivers emergency medicines in flood-affected regions. The system must determine the safest and shortest route despite blocked roads, changing weather conditions, and varying travel costs. The solution should minimize delivery time while ensuring safe and efficient navigation.

2. SMART CITY TRAFFIC SIGNAL OPTIMIZATION

Problem Statement

Develop an AI-based traffic signal optimization system that minimizes traffic congestion, vehicle waiting time, and fuel consumption by dynamically adjusting traffic signal timings according to real-time traffic conditions. The system should efficiently handle a large search space and avoid getting trapped in locally optimal solutions.

3.AUTONOMOUS MARS ROVER NAVIGATION

Problem Statement

Develop an AI-based autonomous rover capable of exploring unknown Martian terrain, collecting samples, avoiding obstacles, and making real-time navigation decisions with limited sensor information. The rover should continuously update its knowledge of the environment to ensure safe and efficient exploration.

4.UNIVERSITY EXAM TIMETABLE GENERATION USING CSP

Problem Statement

Develop an AI-based university exam timetable generation system that schedules exams without conflicts while satisfying multiple constraints such as limited exam halls, faculty availability, subject precedence, and special accommodations for students. The system should efficiently generate a valid timetable that is scalable for a large number of courses and students.

5.STRATEGIC GAME AI USING MINIMAX AND ALPHA-BETA PRUNING

Problem Statement

Develop an AI-based game-playing system for a real-time strategy game that makes intelligent decisions within strict time limits. The system must evaluate multiple possible game states, efficiently search a large decision tree using Minimax with Alpha-Beta Pruning, and select the best move while maintaining optimal performance in real time.